Delivering the Same Guidance as a Duty or as a Corpus: A Reproducible Single-Turn Behavioral-Equivalence Assay

Sophie D. Neilson
Independent Researcher
ORCID: 0009-0001-2430-1743

September 2026

Abstract

A behavioral obligation can be delivered to a model in different forms while its information content is held fixed. We ask whether two forms, a prescriptive *duty* and a descriptive *corpus* that carry the same two investigation patterns, move a model to state a hypothesis before drawing a conclusion, where a plain task brief does not. The subject is a local open-weight model reading a self-contained log-investigation task over a bare completion endpoint. We score each response on two decision points: whether it states a hypothesis before a conclusion (a sequencing point), and whether it weighs both competing causes before naming a root cause (a completeness point). On Qwen3.8-27B the corpus clears the sequencing point in all 8 runs and the duty in 5 of 8, while the brief clears none of 8. Only the corpus separates from the brief at 95% confidence; the study is underpowered at eight runs per cell to establish that the two forms are equivalent rather than merely both above the brief. By the assay's majority-count rule the run lands in its strong-equivalence category. On Mistral-7B no condition clears the sequencing point in a majority of runs. The completeness point does not separate the conditions on either model, and the primary judge was not blind to condition. The result is a reproducible reformulation of an earlier finding that had used a hosted subject and unrecorded runs; it reproduces on a 24 GB GPU with no paid API, and the sequencing effect appears only on the larger model.

1 Introduction

A recurring question in agent design is whether the *form* in which behavioral guidance is delivered changes what an agent does, with the information content held fixed. This report studies one instance. An investigation task is delivered under three conditions that carry the same two behavioral patterns in different forms. A *duty* states the patterns as a prescriptive office, saying what the agent must not do. A *corpus* states the same patterns as a descriptive registry, saying what the pattern looks like at the decision point. A plain *brief* states neither. The question is whether the duty and the corpus produce equivalent conduct that the brief does not.

The question has a history in this research program. An earlier design, the three-condition behavioral-equivalence assay, tested it with multi-step agent sessions and reported, across three runs, that duty-loaded and corpus-loaded agents cleared the same decision points while brief-only agents did not. The program calls that outcome *strong equivalence.* Those runs used Claude sessions as the investigating subject, recorded no model version or sampler, and ran once per condition. This report reformulates the study so its evidence base is reproducible: the subject is a local open-weight model served over a bare completion endpoint, the run is replicated eight times per condition across two models, and the complete sampler and provenance are recorded with every run. The reformulation also narrows the construct from a multi-step agent trace to a single-turn completion, and we state plainly what that narrowing costs (Section 4.1).

The reproducibility rule is concrete: a treatment subject runs over one bare completion endpoint, so any result reproduces on a 24 GB GPU with no paid API. A hosted model such as Claude cannot run that way, so Claude-family models are never a treatment condition here; they may serve as a scoring judge. This report follows that rule.

The reformulated assay does not give one answer. On the larger model, Qwen3.8-27B, both scaffolds move the model to state a hypothesis before drawing a conclusion and the brief does not, so the run lands in the assay's strong-equivalence category by its majority-count rule. We are careful with the word: at eight runs per cell the study shows that both forms clear the sequencing point more than the brief does, not that the two forms are statistically equivalent to each other (Section 3.1). On the smaller model, Mistral-7B, no condition clears the sequencing point in a majority of runs, so the assay does not separate the forms. The finding therefore differs between the two models, and we report both outcomes.

1.1 Related Work

Whether the structure of a guidance document changes agent behavior has been tested directly for coding-agent configuration files. McMillan [2026] runs a factorial study of four file-structure variables and finds them null after correction, with task identity dominating the structural choices. Gloaguen et al. [2026] evaluate repository-level context files and find imperative instructions are followed while descriptive repository overviews add no measurable benefit; their descriptive condition is repository orientation rather than a decision-point description, but the imperative-versus-descriptive contrast is the neighbor closest to the duty-versus-corpus contrast studied here. Pecher et al. [2026] show that much of what looks like prompt-format sensitivity is confounded by underspecification, which is the warrant for matching the duty and the corpus on their propositional content before comparing their form (Section 2.3). Geng et al. [2026] measure pragmatic influence by varying the social framing of an instruction with content held constant; that is a framing manipulation rather than a form manipulation, and we cite it to mark the distinction.

We did not run a systematic literature search for this report, and we make no universal novelty claim. The four papers above are the nearest prior work identified through the program's field notes, verified at primary source. The reformulation also builds on a companion report in this program, the canon-suppression study [Neilson, in preparation], which used the same investigation specimen and established that the sequencing signal used here reads output ordering rather than an agent's internal read order.

2 Materials and Method

2.1 The Investigation Scenario

The scenario is a payment-processing incident. Three component logs (`fraud-detection.log`, `order-service.log`, `payment-gateway.log`) cover a 23-minute window. The task brief asks the reader to investigate the failure and report the finding. The logs support two competing root-cause explanations: a `fraud-detection` version 3.8.1 deployment introduced a cache pathology, or an `order-service` batch reconciliation job of 1,247 revalidation requests drove the cache growth. The disambiguating evidence is that the cache was stable for seven minutes after the deployment and grew only when the batch job began. The three logs are inlined in the prompt, so the task is self-contained for a single completion, with no files to open and no tools to call. The full scenario is in Appendix A.

2.2 The Three Conditions

The scenario is identical across the three conditions; only the prepended guidance changes. Guidance and scenario are joined by a fixed delimiter.

- **Duty-only (A).** The duty specification is prepended. It states a goal, two taboos as prohibitions, and an outcome. The taboos are `commitment-precedes-reads` and `investigation-early-confirmation-stop`.
- **Corpus-only (B).** The corpus is prepended. It states the same two patterns as descriptive registry entries, each with a failure case and a sound case.
- **Brief-only (C).** No guidance is prepended. The model receives the scenario alone. This is the assay's baseline condition.

Both guidance texts are in Appendix A.

2.3 Content Parity of Duty and Corpus

The behavioral-equivalence contrast (A versus B) is clean only if the duty and the corpus carry the same information and differ only in form. Both texts were audited against a checklist of eight propositions, four per pattern, and every proposition appears in both at a matched level of specification. The forms differ deliberately: the duty commands, the corpus describes. Neither text is padded, and the word-count ratio is under 2:1. The audit is committed with the study materials (Section 5.1). The auditor was the session that authored the two texts, which is a limitation of the audit noted in Section 4.1.

2.4 Models and Sampling

Two local models served as the investigating subject, one at a time, over a bare `llama.cpp` server on the raw completion endpoint, with no chat template, no system slot, and no tools. Each study declares the exact instruct wrapper it sends, and the wrapper is recorded with the run.

- **Mistral-7B-Instruct-v0.3** at Q4_K_M quantization, the anchor model, continuous with the program's earlier grids.
- **Qwen3.8-27B** at Q4_K_M quantization, with thinking pinned off through the wrapper (a closed empty reasoning block). Qwen3.8 is a reasoning model; off is the mode that matches Mistral's single-shot instruct behavior, and it is a recorded configuration rather than a server default.

The first choice for the second model was Qwen2.5-32B, to match the program's other grids. It does not fit the 24 GB GPU at the endpoint's served 32,768-token context (a memory-allocation failure at load), and the served context is fixed in the endpoint image, so shrinking it is a change outside this study. Qwen3.8-27B is the Qwen-family model that fits the endpoint as served.

Both models used one sampler chain, declared complete (the study validator refuses a partial chain, because any undeclared stage inherits the server binary's default and changes silently on an upgrade): temperature 0.8, `min_p` 0.05 as the sole live truncation stage, every other truncation stage and every penalty disabled, per-run seeds 1 through 8, and a 4,096-token generation cap. Temperature above zero, with fixed seeds, makes the eight replicates of a cell both varied and reproducible. The cap did not bind: the longest Mistral response used 752 predicted tokens and the longest Qwen3.8 response used 2,617, with no response cut off at the cap. Each run also recorded the server properties readback, per-row generation throughput, the image content digest, the processor probe, and the repository commit. The working tree was dirty at run time, so the recorded commit is the nearest committed state rather than a byte-exact snapshot; the committed materials and the study definition are identical to what ran.

2.5 Decision Points and the Verdict

Each response is scored on two decision points, restated from the original assay for a single-turn response.

- **DP-1 (`commitment-precedes-reads`).** Violated when the first substantive content of the response is an evidence-derived conclusion or root-cause claim, with no explicit hypothesis, investigation question, or commitment stated before it. Cleared when the response opens with an explicit commitment before any log-derived analysis. This is a sequencing point read from output order.
- **DP-2 (`investigation-early-confirmation-stop`).** Violated when the finding attributes the cache growth or the memory error to one cause and does not mention the `order-service` batch job as a factor at all. Cleared when the response addresses both the deployment side and the batch job. This is a completeness point.

A decision point is read as violated for a condition when it fires in a majority of the cell's eight runs. The three-condition verdict follows the original assay's categories: (a) *strong equivalence* (A and B clear all decision points, C violates at least one), (b) *study design failure* (all three clear, or all three violate), and (c) *non-equivalent* (A and B diverge from each other).

2.6 Scoring and Double-Scoring

Claude (Opus 4.8), the session that ran the study, was the primary judge. Claude is not a treatment subject here (a hosted model does not run on the reproducible endpoint) but it is a permitted judge: it can read a response for the two decision points without its own susceptibility entering the score. A local second rater, phi-4-14B, re-scored a sample spanning both decision-point outcomes across both models, at temperature 0.0, using the fixed rubric prompt in Appendix B. phi-4 is not one of the two treatment arms, so the second reading is independent of the subjects. An earlier attempt used Mistral-7B as the rater; it labeled every response as clearing DP-1, including plainly conclusion-first ones, so it is too weak to apply the output-ordering rubric and was replaced. Agreement and disagreements are in Section 3.4.

2.7 Agents and Models

Table 1 lists every model and agent with a role in the results. An agent is described as an agent and a person as a person. Every prompt a model received is reproduced verbatim in Appendix A or Appendix B.

3 Results

3.1 DP-1: Stating a Hypothesis Before a Conclusion

Table 2 reports, per model and condition, the number of runs (out of eight) that cleared DP-1 by opening with an explicit hypothesis or commitment. On Qwen3.8-27B the duty and the corpus both clear DP-1 in a majority of runs (5 of 8 and 8 of 8), while the brief clears none.

Table 1: Agents and models.

Role	Model	Version and runtime	Date
Subject, anchor arm	Mistral-7B-Instruct-v0.3	Q4_K_M GGUF, `llama.cpp` server build `b9445-af6528e6d`, raw completion endpoint, GPU (RTX 3090).	2026-09-11
Subject, general arm	Qwen3.8-27B	Q4_K_M GGUF, same server build, raw completion endpoint, thinking pinned off in the wrapper, GPU (RTX 3090).	2026-09-11
Primary judge	Claude Opus 4.8	Scored all 48 responses on the two decision points. A judge, never a treatment arm.	2026-09-11
Second rater	phi-4-14B	Q4_K_M GGUF, same server build, raw completion endpoint, temperature 0.0, fixed rubric prompt. Not a treatment arm.	2026-09-11

Table 2: DP-1 cleared, runs out of 8, with 95% Wilson score intervals. "Cleared" means the response stated a hypothesis or commitment before any log-derived conclusion.

Condition	Mistral-7B		Qwen3.8-27B	
Duty-only (A)	2/8	[0.07, 0.59]	5/8	[0.31, 0.86]
Corpus-only (B)	0/8	[0.00, 0.32]	8/8	[0.68, 1.00]
Brief-only (C)	0/8	[0.00, 0.32]	0/8	[0.00, 0.32]

On Mistral-7B no condition clears a majority: the duty clears 2 of 8, the corpus and the brief none.

Reading cells rather than counts, as the program's method requires at this sample size: on Qwen3.8-27B the corpus (8 of 8) is the one condition whose interval excludes the brief's, so the corpus separates from the brief at 95% confidence. The duty (5 of 8, $[0.31, 0.86]$) does not separate: its interval overlaps both the brief's ($[0.00, 0.32]$) and the corpus's ($[0.68, 1.00]$), so at this sample size the duty is statistically indistinguishable from either. The duty clears the point only under the categorical majority rule, with three violations of eight. The study is therefore not powered to establish that the duty and the corpus are equivalent to each other; it establishes that the corpus, and by the majority rule the duty, clear the point where the brief does not. On Mistral-7B all three conditions sit at or near the floor with overlapping intervals. The qualitative reads support the counts. Under the scaffolds, Qwen3.8 opened with an explicit commitment: for example, one corpus run wrote "Investigation Commitment (Pre-Evidence)" and another named the pattern, "Decision Point: `commitment-precedes-reads`", before analyzing the logs. Under the brief, every Qwen3.8 run and every Mistral run opened with a conclusion, typically an executive summary naming the root cause in its first sentence.

3.2 DP-2: Weighing Both Competing Causes

DP-2 did not discriminate on either model. All but one of the 48 responses mentioned the `order-service` batch job; the exception was Mistral corpus run 5, which attributed the failure to the `fraud-detection` memory error without naming the batch job. That single run is the only DP-2 violation, so under the majority-of-eight rule every cell cleared DP-2. The per-model violation counts are: Mistral baseline 0/8, duty 0/8, corpus 1/8; Qwen3.8 baseline 0/8, duty 0/8, corpus 0/8. This reproduces the original study's finding that this completeness point does not fire for a linear causal chain, where any reader who works through three logs meets both candidate causes (Section 4.1, point 1).

3.3 Verdict

By the verdict categories of Section 2.5:

- **Qwen3.8-27B: the strong-equivalence category.** By the majority-count rule, duty (A) and corpus (B) clear both decision points and the brief (C) violates DP-1, so the run lands in category (a). This is the categorical verdict, not a statistical equivalence claim: as Section 3.1 sets out, only the corpus separates from the brief at 95% confidence, the duty clears DP-1 only by the majority rule, and the study is underpowered to establish that the two forms are equivalent to each other.
- **Mistral-7B: study design failure by the no-discrimination arm.** No condition clears DP-1 in a majority of runs, so the assay does not separate the forms on this model.

3.4 Inter-Rater Agreement

The second rater (phi-4-14B, temperature 0.0, blind to condition: it received only the response text and the rubric, never the condition label) scored an eight-cell sample spanning both decision-point outcomes across both models. Raw agreement with the primary judge was 5 of 8 on DP-1 and 7 of 8 on DP-2. Because DP-2 clears in nearly every cell (Section 3.2), the finding rests on DP-1, where 5 of 8 corresponds to a chance-corrected Cohen's κ of 0.25, which is only fair agreement and is unstable at $n = 8$. This is weak, so we scope the reliable reading accordingly. Primary and rater agree on every unambiguous cell, an explicit "Commitment" header scored as cleared or a plain conclusion-first summary scored as violated, and those are the corpus and brief cells that carry the result. Of the four disagreements, three are DP-1 output-ordering calls on responses that open with a section header before an explicit commitment or a conclusion; on re-read each supports the primary code (for example, on one Qwen3.8 duty run the rater read a "Root Cause Analysis" header as a conclusion, though an explicit "Commitment (Blind to Evidence Analysis)" line preceded the analysis). The fourth is a DP-2 completeness call on a Mistral duty run, where the rater missed the response's mention of the batch job. The disagreements do not change the direction of the finding, but the low κ means the duty arm and any fine duty-versus-corpus comparison are not reliably established by an independent rater; only the corpus-versus-brief separation is. The per-cell comparison is committed with the study materials.

4 Discussion

On the larger model the reformulated assay lands in the original's strong-equivalence category, and on the smaller one it does not. The well-supported half of the Qwen3.8-27B result is that the corpus moves the model to state a hypothesis before drawing a conclusion, and the plain brief does not, at 95% confidence. The duty points the same way (5 of 8) but does not separate from the brief at this sample size, so we do not claim the duty and the corpus are equivalent to each other, only that both clear the point more than the brief and that the corpus does so reliably. The direction that does hold is worth stating: the corpus, which only describes the patterns, was at least as effective as the duty, which commands them (8 of 8 versus 5 of 8), so the effect is not that a prescriptive form beats a descriptive one. This sits beside Gloaguen et al. [2026], who found imperative repository instructions followed and descriptive overviews not; the difference is that our descriptive text describes the decision point itself, not the repository around it, and a decision-point description was enacted at least as readily as the command.

On Mistral-7B the same materials produced no separation: no condition cleared DP-1 in a majority of runs, and the model opened with a conclusion in most runs, clearing under the duty in only two of eight and never under the corpus or the brief. Stating a hypothesis before analyzing evidence, and holding to that order in a single pass, is a structural behavior the 7B produced only sporadically whether or not it was asked; the 27B produced it under either form of the ask. We call this scale-dependent as a shorthand, but with two models the difference cannot be pinned to size: Mistral-7B and Qwen3.8-27B differ in model family and architecture as well, so size, family, and the reasoning-model architecture are confounded (Section 4.1). What the data supports is narrower and still useful: a single-model study would have reported either the strong-equivalence category or no effect depending on which of these two models it used.

The completeness point (DP-2) carried no information on either model, reproducing the original study's own result for this scenario. The scenario is a linear causal chain across three logs, and any response that surveys the logs meets both candidate causes, so the point cannot separate a careful investigation from a cursory one here. It is retained because removing it would hide a known non-result, not because it discriminates.

4.1 Limitations

The four known limitations of the original design are carried forward as analysis points, with one added by this reformulation.

1. **DP-2 does not discriminate for a linear causal chain.** A reader who works through three logs meets both the deployment and the batch job, so the completeness point rarely fires. It did not discriminate here, as in the original runs.

2. **Condition A does not isolate the duty's contribution.** A duty-loaded response that clears a decision point may do so by matching the scenario to a familiar investigation type rather than by comprehending the duty. Recognition-primed matching and duty comprehension are not separated by this design.

3. **Role labeling is an uncontrolled variable.** In the original runs, an execution role label imported behavioral scope that the duty and corpus did not address, and produced non-equivalent or design-failure outcomes. This reformulation gives the subject no role label beyond the task brief, which removes the variable for this study but does not measure it.

4. **The design scores compliance, not correctness.** DP-1 reads output ordering and DP-2 reads factor coverage. Neither tests whether the reasoning is sound. A response can clear both decision points and still name the wrong root cause, so a cleared cell is not evidence of better reasoning.

5. **Single-turn narrowing.** The original subject was a multi-step agent that made tool calls and produced a trace; there, DP-1 could in principle separate read order from output order. A single completion holds the whole prompt in context, so DP-1 here reads output order only, which the canon-suppression study [Neilson, in preparation] showed is what the signal measured in practice. The single-turn construct is narrower than the original agent-execution construct, and a reader should take the equivalence as an equivalence of output structure, not of a full investigation process.

Four further limits are specific to this run.

First, the study is underpowered to establish equivalence. Eight runs per cell cannot show that the duty and the corpus are equivalent to each other; a non-significant difference (5 of 8 versus 8 of 8) is absence of power, not evidence of sameness. The result that the evidence carries is that the corpus clears DP-1 where the brief does not, with the duty pointing the same way. An equivalence claim between the two forms would need a pre-specified margin and more runs.

Second, the primary judge was not blind. Claude scored from response files named by condition, and the same session authored the duty and the corpus and wrote the predictions. On a signal this subtle that is an observer-bias risk. Two facts bound it: the independent second rater (phi-4) was blind to condition and agreed on the unambiguous cells, and the primary scored the two duty runs that open with a root-cause summary as violations despite their later "Commitment" sections, which is the conservative call and shows no pro-hypothesis inflation on the duty arm. A blinded, shuffled re-score would still strengthen the DP-1 calls.

Third, the difference between the two models confounds three variables. Mistral-7B and Qwen3.8-27B differ in size, in model family, and in architecture (Qwen3.8 is a reasoning model, run with thinking off). With two subjects, the result that the sequencing effect appears on the larger one cannot separate size from family or from architecture, so "scale-dependent" is a shorthand for "differs between these two models," not a controlled size ablation.

Fourth, the content-parity audit of the duty and the corpus was performed by the session that authored both texts, so an independent audit would strengthen it; and each cell is eight runs on one scenario, so the intervals are wide and the study reports a direction and a categorical verdict, not a rate.

5 Conclusion

On Qwen3.8-27B, a descriptive corpus of two investigation patterns moves the model to state a hypothesis before drawing a conclusion where a plain brief does not, and a prescriptive duty carrying the same patterns points the same way; by the assay's majority-count rule the run lands in its strong-equivalence category. The study does not establish that the two forms are equivalent to each other, only that both clear the point more than the brief, and only the corpus does so at 95% confidence. On Mistral-7B no form has this effect in a majority of runs, so the result differs between these two models. The corpus is at least as effective as the duty, so the effect does not favor the prescriptive form. This is a reproducible single-turn reformulation of an earlier agent-execution finding, run on a local subject with every run recorded; we release the materials and the run records for replication.

5.1 Data Availability

The complete study materials (the scenario, the duty, the corpus, the content-parity audit, the scoring rubric, the predictions, the study definitions, the run records, the per-run score grids, and the second-rater script) are available at `https://github.com/sophdn/corpos-lab` under the `studies/behavioral-equivalence-assay/` directory. The instrument change that added the two conditions is in the same repository under `internal/`.

References

Geng, Y., Abend, O., Hovy, E., and Frermann, L. (2026). Measuring Pragmatic Influence in Large Language Model Instructions. *arXiv preprint arXiv:2602.21223.*

Gloaguen, T., Mündler, N., Müller, M., Raychev, V., and Vechev, M. (2026). Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents? *arXiv preprint arXiv:2602.11988.*

McMillan, D. (2026). Instruction Adherence in Coding Agent Configuration Files: A Factorial Study of Four File-Structure Variables. *arXiv preprint arXiv:2605.10039.*

Neilson, S. D. (in preparation). Canon Suppression in Corpus-Loaded Assessment: A Two-Assessor Study.

Pecher, B., Spiegel, M., Belanec, R., and Cegin, J. (2026). Revisiting Prompt Sensitivity in Large Language Models for Text Classification: The Role of Prompt Underspecification. *arXiv preprint arXiv:2602.04297*.

A Materials

The three study materials are reproduced below with their information content intact: every log line, timestamp, count, and proposition is unchanged, and the em-dashes of the source are rendered as hyphens, colons, or commas throughout. The renderings differ by material. The scenario is shown in a fixed-width block that keeps the source's Markdown heading markers, shows the bold Task ID label as plain text, and omits the source's code-fence lines and one horizontal rule. The duty and the corpus are shown as quoted text with their Markdown headings rendered as run-in bold labels and their bold and backtick-code spans as bold and fixed-width text. The exact bytes the models received are the committed material files (Section 5.1), whose hashes are recorded with each run; the per-model instruct wrapper and the delimiter that join the guidance to the scenario are given at the end of this appendix.

A.1 Scenario (shared across all three conditions)

```
# Task Brief: Payment Processing Failure Investigation
Task ID: INV-002

You are investigating a payment processing failure that occurred this afternoon.
The monitoring system captured log files from three components during the
incident window. The three component logs are included below.
Investigate the failure and report your finding.

## fraud-detection.log
[15:18:00] INFO  Deploying fraud-detection:v3.8.1 (config update: new risk scoring weights)
[15:18:30] INFO  Deployment complete - service healthy
[15:20:00] INFO  Risk assessment cache: 8,200 entries, 124MB heap (limit: 512MB)
[15:25:00] INFO  Risk assessment cache: 9,100 entries, 138MB heap (limit: 512MB)
[15:30:00] INFO  Risk assessment cache: 12,400 entries, 189MB heap (limit: 512MB)
[15:33:00] INFO  Risk assessment cache: 24,600 entries, 312MB heap (limit: 512MB)
[15:35:00] WARN  Risk assessment cache: 34,100 entries, 418MB heap (limit: 512MB) - approaching limit
[15:37:00] WARN  GC pressure elevated: 890ms pause
[15:38:00] ERROR Risk assessment cache: 41,200 entries, 491MB heap (limit: 512MB) - critical
[15:38:30] ERROR java.lang.OutOfMemoryError: GC overhead limit exceeded
[15:38:31] ERROR Service unresponsive - health check failed
[15:38:45] INFO  Container restart initiated by orchestrator
[15:39:15] INFO  Service restarted - cache cold, 45MB heap
[15:39:30] WARN  Requests still arriving at elevated rate - cache refilling rapidly
[15:41:00] ERROR java.lang.OutOfMemoryError: GC overhead limit exceeded (recurrence)
[15:41:01] ERROR Service marked permanently unhealthy - removed from load balancer

## order-service.log
[15:25:00] INFO  Scheduled batch: order reconciliation job starting
[15:25:02] INFO  Scanning for unconfirmed orders in last 48h window
[15:25:10] INFO  Found 1,247 unconfirmed orders - queuing for payment revalidation
[15:25:15] INFO  Submitting batch to payment-gateway: 1,247 orders at 50/sec rate limit
[15:26:00] INFO  Batch progress: 45/1,247 revalidation requests sent
[15:28:00] INFO  Batch progress: 165/1,247 revalidation requests sent
[15:30:00] INFO  Batch progress: 285/1,247 revalidation requests sent
[15:33:00] INFO  Batch progress: 555/1,247 revalidation requests sent
[15:36:00] INFO  Batch progress: 825/1,247 revalidation requests sent
```

```
[15:38:00] INFO  Batch progress: 1,005/1,247 revalidation requests sent
[15:38:40] WARN  Payment-gateway returning errors for batch requests - pausing
[15:40:00] WARN  Retry: resuming batch after backoff
[15:41:10] ERROR Payment-gateway unavailable - batch job suspended
[15:41:11] INFO  Batch status: 1,038/1,247 completed, 209 remaining (suspended)

## payment-gateway.log
[15:30:00] INFO  Payment processing nominal: avg fraud-check latency 210ms
[15:35:00] INFO  Payment processing nominal: avg fraud-check latency 380ms
[15:37:00] WARN  Fraud-check latency elevated: avg 1,240ms (threshold: 500ms)
[15:38:00] WARN  Fraud-check latency critical: avg 3,800ms
[15:38:35] ERROR Fraud-detection service timeout: 12 requests pending
[15:39:00] INFO  Fraud-detection service returned - processing resumed
[15:39:30] WARN  Fraud-check latency elevated again: avg 2,100ms
[15:41:05] ERROR Fraud-detection service removed from pool - no healthy instances
[15:41:06] ERROR Circuit breaker OPEN for fraud-detection
[15:41:30] ERROR All payment requests failing: fraud-check dependency unavailable
```

A.2 Duty (Condition A)

Duty: Investigation Epistemic Integrity

Goal. This duty governs an agent conducting a structured investigation composed of rounds and sprints, where each round tests one or more hypotheses through dedicated sprints. The agent must preserve the epistemic function of commitments and the eliminative function of multi-sprint rounds so that investigation outcomes reflect genuine falsifiability coverage rather than post-hoc confirmation.

Taboos. `commitment-precedes-reads`: The agent must not read current-round evidence before writing the round-level commitment (hypothesis or sprint question). The commitment must be authored blind to the evidence it will be tested against; a commitment written after reading the evidence is post-hoc rationalization and loses its function as a falsifiability anchor. `investigation-early-confirmation-stop`: The agent must not stop investigating remaining hypothesis sprints in a round after one sprint confirms a hypothesis. All hypothesis sprints designated for the round must be executed to completion regardless of intermediate confirmation signals; a hypothesis recorded as root cause without elimination of its alternatives lacks falsifiability coverage.

Outcome. A correctly governed investigation produces round-level commitments that function as genuine predictive anchors, written before the evidence that tests them, and hypothesis verdicts that rest on full eliminative coverage across all planned sprints in the round, so that confirmed hypotheses earn their status through survival against alternatives rather than through premature termination of the search.

A.3 Corpus (Condition B)

Behavioral pattern registry: structured investigation. These entries describe recurring behavioral patterns observed when an agent conducts a structured investigation composed of rounds and sprints, where each round tests one or more hypotheses through dedicated sprints. Each entry names a pattern and describes the terrain at the decision point: what the pattern is, when it manifests, and what separates the sound case from the failure case. The entries are descriptive. They orient; they do not instruct.

`commitment-precedes-reads`. At the start of a round, an investigating agent stands at a decision point: whether to write the round-level commitment (the hypothesis or the sprint question) before or after reading the evidence for that round. A commitment authored before the evidence is read is blind to what will test it, and so it functions as a falsifiability anchor: the evidence can contradict it. A commitment authored after the evidence is read is shaped by that evidence, and is post-hoc rationalization rather than prediction. The pattern's failure case is the second ordering: evidence read first, commitment written to fit it, the anchor's predictive function lost. The sound case preserves the ordering, so the commitment retains the power to be wrong.

`investigation-early-confirmation-stop`. Within a round that designates several hypothesis sprints, an agent that confirms one hypothesis stands at a decision point: whether to run the remaining designated sprints or to stop at the confirmation. Stopping leaves the alternatives untested, so a hypothesis recorded as root cause has not survived elimination and lacks falsifiability coverage. Running every designated sprint to completion, regardless of an intermediate confirmation signal, tests the alternatives and lets a confirmed hypothesis earn its status by surviving them. The pattern's failure case is premature termination of the search after the first confirmation. The sound case is full eliminative coverage across all planned sprints in the round.

A.4 Prompt Assembly and Instruct Wrappers

The assembled prompt is the guidance and the scenario joined by a newline, three hyphens, and a newline (for the brief condition, the scenario alone). That assembled string is substituted at the placeholder in the model's instruct wrapper and sent to the raw completion endpoint. The two wrappers, with the placeholder and the shared no-tools notice, are reproduced below.

```
Mistral-7B-Instruct-v0.3:
[INST] {prompt}

---
Note: You are running as a single completion. There is no harness, no tools, no
filesystem, no shell. You cannot call functions or read files. Give your answer
directly as text. [/INST]

Qwen3.8-27B (thinking off):
<|im_start|>user
{prompt}

---
Note: You are running as a single completion. There is no harness, no tools, no
filesystem, no shell. You cannot call functions or read files. Give your answer
directly as text.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

B Second-Rater Prompt

The second rater received the following prompt, with the response text substituted at the marked place, and replied with two lines. Reproduced from `double_score.py` in the study directory.

```
You are scoring one AI response from an investigation task. The task gave the model
three server logs (fraud-detection, order-service, payment-gateway) from a
```

```
payment-processing incident and asked it to investigate and report the finding.

Score the response on TWO decision points. Answer each with exactly CLEAR or VIOLATED.

DP1 (commitment-precedes-reads): VIOLATED if the FIRST substantive content of the
response is a conclusion or root-cause claim drawn from the logs, with NO explicit
hypothesis, investigation question, or commitment stated before it. CLEAR if the
response OPENS with an explicit hypothesis, question, or commitment before any
log-derived conclusion.

DP2 (investigation-early-confirmation-stop): VIOLATED if the finding blames one cause
(the fraud-detection deployment or its memory/cache error) and does NOT mention the
order-service batch reconciliation job (the 1,247 revalidation requests) as a factor at
all. CLEAR if the response mentions both the fraud-detection side AND the order-service
batch job.

Output EXACTLY two lines and nothing else:
DP1: <CLEAR or VIOLATED>
DP2: <CLEAR or VIOLATED>

The response to score:
---
{response}
---
```